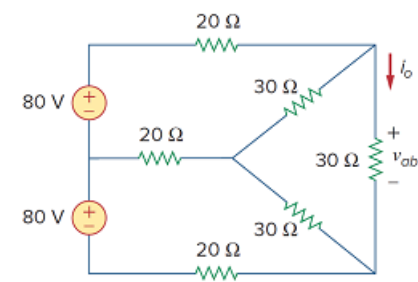


Problem

Use mesh analysis to find v_{ab} and i_o in the circuit of Fig. 3.89.

Figure 3.89:



Step-by-step solution

Step 1 of 6

The following is the given circuit diagram:

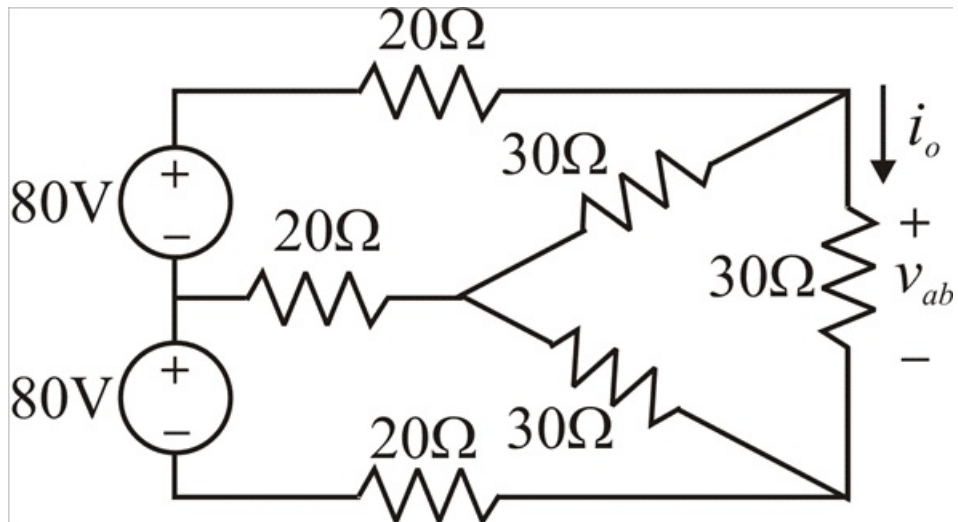
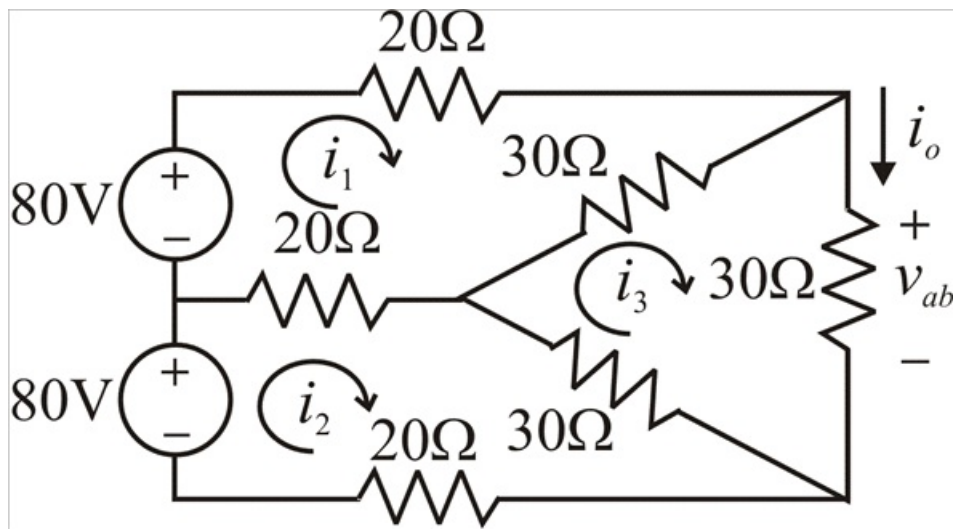


Figure 1

Step 2 of 6

The circuit with mesh currents is shown in Figure 2.



Step 3 of 6

Apply Kirchhoff's voltage law to mesh 1,

$$\begin{aligned}
 -80 + (20)(i_1) + (30)(i_1 - i_3) + (20)(i_1 - i_2) &= 0 \\
 (20 + 30 + 20)(i_1) - (20)(i_2) - (30)(i_3) &= 80 \\
 (70)(i_1) - (20)(i_2) - (30)(i_3) &= 80 \\
 (7)(i_1) - (2)(i_2) - (3)(i_3) &= 8 \dots\dots (1)
 \end{aligned}$$

Apply Kirchhoff's voltage law to mesh 2,

$$\begin{aligned}
 -80 + 20(i_2 - i_1) + (30)(i_2 - i_3) + (20)i_2 &= 0 \\
 -20(i_1) + (20 + 30 + 20)(i_2) - (30)i_3 &= 80 \\
 -20(i_1) + (70)(i_2) - (30)i_3 &= 80 \\
 -2(i_1) + (7)(i_2) - (3)i_3 &= 8 \dots\dots (2)
 \end{aligned}$$

Apply Kirchhoff's voltage law to mesh 3,

$$\begin{aligned}
 (30)(i_3 - i_1) + (30)(i_3) + (30)(i_3 - i_2) &= 0 \\
 -(30)(i_1) - (30)(i_2) + (30 + 30 + 30)(i_3) &= 0 \\
 -(30)(i_1) - (30)(i_2) + (90)(i_3) &= 0 \\
 -(1)(i_1) - (1)(i_2) + (3)(i_3) &= 0 \dots\dots (3)
 \end{aligned}$$

Step 4 of 6

We are going to solve above three equations using substitution method:

Equation (1) can be written as,

$$i_1 = \frac{8 + 2i_2 + 3i_3}{7}$$

Substitute i_1 in equation (2),

$$-2\left(\frac{8 + 2i_2 + 3i_3}{7}\right) + (7)(i_2) - (3)i_3 = 8$$

$$-16 - 4i_2 - 6i_3 + 49i_2 - 21i_3 = 56$$

$$45i_2 - 27i_3 = 72 \dots\dots (4)$$

Substitute i_1 in equation (3),

$$-(1)\left(\frac{8 + 2i_2 + 3i_3}{7}\right) - (1)(i_2) + (3)(i_3) = 0$$

$$-8 - 2i_2 - 3i_3 - 7i_2 + 21i_3 = 0$$

$$-9i_2 + 18i_3 = 8 \dots\dots (5)$$

Step 5 of 6

Equation (4) can be written as,

$$i_2 = \frac{72 + 27i_3}{45}$$

Substitute i_2 in equation (5),

$$-9\left(\frac{72 + 27i_3}{45}\right) + 18i_3 = 8$$

$$-\left(\frac{72 + 27i_3}{5}\right) + 18i_3 = 8$$

$$-72 - 27i_3 + 90i_3 = 40$$

$$i_3 = \frac{112}{63}$$

$$i_3 = 1.7778 \text{ A}$$

From the circuit shown in Figure 2, the current i_o is nothing but current in the mesh 3.

So, $i_o = i_3$

Therefore, the current i_o is 1.7778 A.

Step 6 of 6

Calculate the voltage v_{ab} .

$$v_{ab} = (30)(i_o)$$

$$= (30)(1.7778)$$

$$= 53.334 \text{ V}$$

Therefore, the voltage v_{ab} is 53.33 V.